WEEKS 5 – 6

Mesh Networking & Off-Grid Communication

LoRa · Meshtastic · MQTT · Resilient Networks

Session 1
LoRa · Meshtastic · Topology

Session 2
MQTT · Bridging · Resilience

Session 1

LoRa Radio Fundamentals & Meshtastic Setup

→ LoRa radio fundamentals

→ Meshtastic setup & configuration

→ Mesh network topology

What is LoRa?

Long Range

Signals can travel 2–15 km in cities up to 40-400 km in open terrain, penetrating walls and terrain obstacles.

Low Power

Devices run for months or years on AA batteries — ideal for remote deployments.

Low Bandwidth

Not for video or voice. Think short text, sensor data, coordinates — ~250 bytes at a time.

License-free bands

868 MHz (EU) / 915 MHz (US) — no ham license required for basic use.

LoRa ≠ LoRaWAN

LoRa: Origin & History

The technology behind every Meshtastic node — and where it came from.

2009

Cycleo founded

A small startup in Grenoble, France develops chirp spread spectrum modulation for low-power radio. The core LoRa IP is born.

2012

Semtech acquires Cycleo

Semtech — a US semiconductor company — buys Cycleo and takes ownership of the LoRa patent. They begin producing the SX127x chip series used in hardware today.

2015

LoRa Alliance & LoRaWAN standardised

Semtech co-founds the LoRa Alliance with IBM, Cisco, and others. They standardise LoRaWAN — a centralised IoT protocol built on top of LoRa radio.

2018

Meshtastic project begins

An open-source project repurposes LoRa for decentralised peer-to-peer mesh networking — no gateways, no cloud, no infrastructure required.

LoRa = the radio modulation (the chirp), owned by Semtech. **LoRaWAN** = centralised IoT protocol on top of LoRa. **Meshtastic** = open, decentralised mesh — same radio, different vision.

LoRa vs. WiFi

Both are wireless — but built for completely different jobs.

	LoRa	WiFi
Range	2 – 400 km	20 – 100 m
Power	Very low (years on battery)	High (hours on battery)
Speed	0.3 – 50 kbps	10 – 10,000 Mbps
Good for	Short texts, GPS, sensors	Video, web, files
Needs router?	No — nodes talk directly	Yes — needs infrastructure
License	None (ISM band)	None (ISM band)

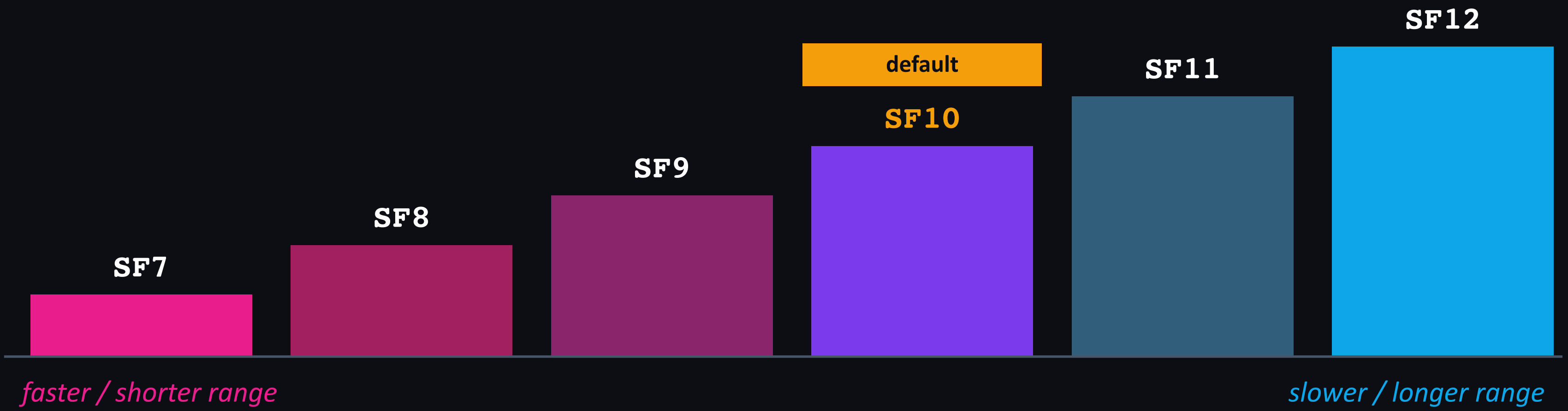
LoRa trades speed for distance and battery life. WiFi trades range for bandwidth. Neither is better — they solve different problems.

The Spreading Factor

LoRa doesn't have a fixed range. You dial the range in — and pay for it in time.

Think of it like shouting vs. whispering slowly.

You can shout one word fast and hope your friend hears it — or you can draw each syllable out slowly and clearly so they catch every sound even from far away. A higher spreading factor is drawing it out. The message takes longer, but it carries further and survives more noise.



each step up =
2× the airtime

each step up =
2× the range

SF10 default =
sweet spot for city-scale mesh

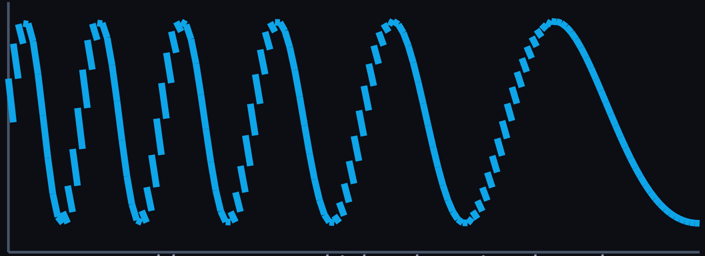
The Chirp: How LoRa Encodes Data

LoRa does not transmit at one fixed frequency — it sweeps continuously across a bandwidth. That sweep is a chirp.

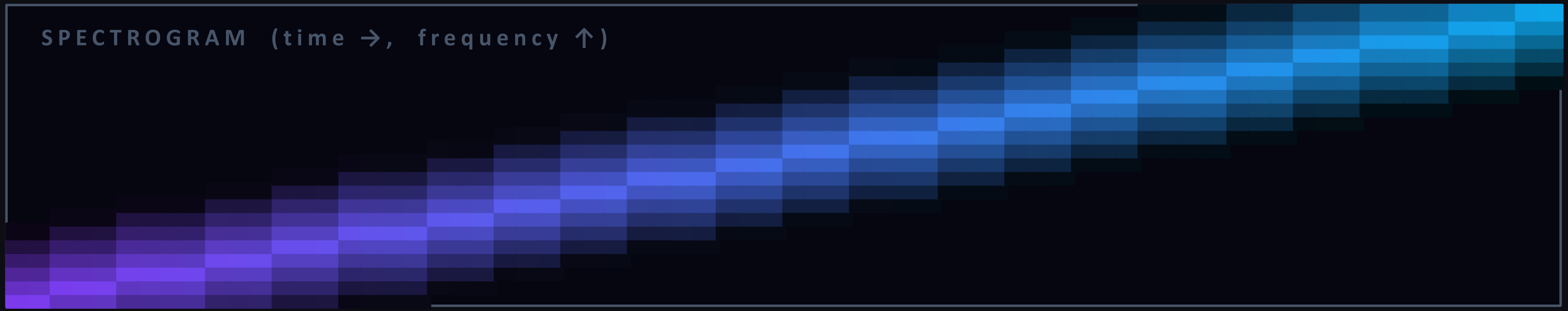
UPCHIRP

frequency sweeps low to high — encodes one symbol

DOWNCHIRP



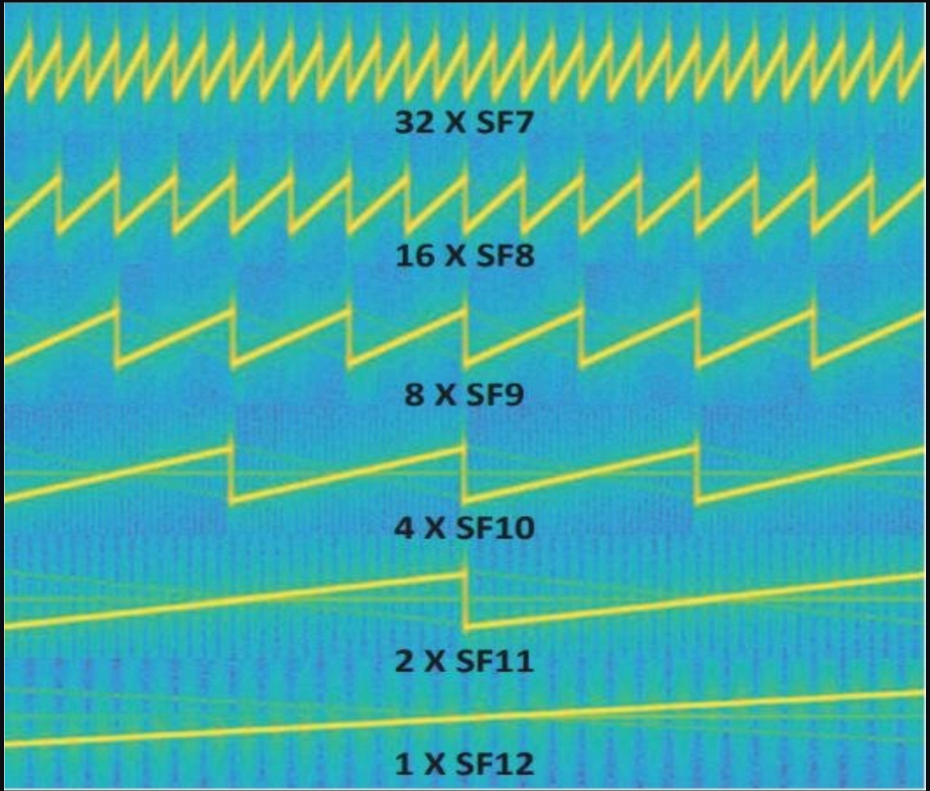
preamble — sweeps high to low, signals packet incoming



The diagonal stripe is the chirp — the receiver locks onto it even buried in noise

What Chirps Actually Look Like

Real SDR waterfall capture — each row is one spreading factor, shown at the same scale.



SF7

32 chirps visible — tiny, fast zigzags

SF8

16 chirps — slightly wider sweeps

SF9

8 chirps — getting visibly slower

SF10

4 chirps — Meshtastic default, clear diagonal

SF11

2 chirps — half the screen per sweep

SF12

1 chirp — spans the full capture width

[Watch: LoRa chirp explained — youtu.be/dxYY097QNs0](https://youtu.be/dxYY097QNs0)

What is Meshtastic?

An open-source, off-grid mesh communication platform built on LoRa radio hardware.

- Works without internet, cell, or WiFi
- End-to-end encrypted by default
- Designed for days of battery life
- GPS position sharing built-in
- Phone app via Bluetooth (iOS & Android)
- Python API for scripting & hacking

Common Hardware

Heltec WiFi LoRa 32
Built-in OLED screen

T-Beam / T-Echo
GPS + LoRa combo

RAK WisBlock
Modular, pro-grade

Seeed SenseCAP T1000
Ultra-compact

DIY ESP32 + SX1276
Breadboard it yourself

Setting Up Your Node

1 Flash firmware

Download Meshtastic flasher → connect USB → select your board → flash

2 Connect via Bluetooth

Open Meshtastic app (iOS or Android) → scan → pair with your node

3 Set your region

Settings → LoRa Config → Region: EU_868 (Berlin) or US_915

4 Set a channel

Default channel = 'LongFast'. Custom channels use a pre-shared key for privacy.

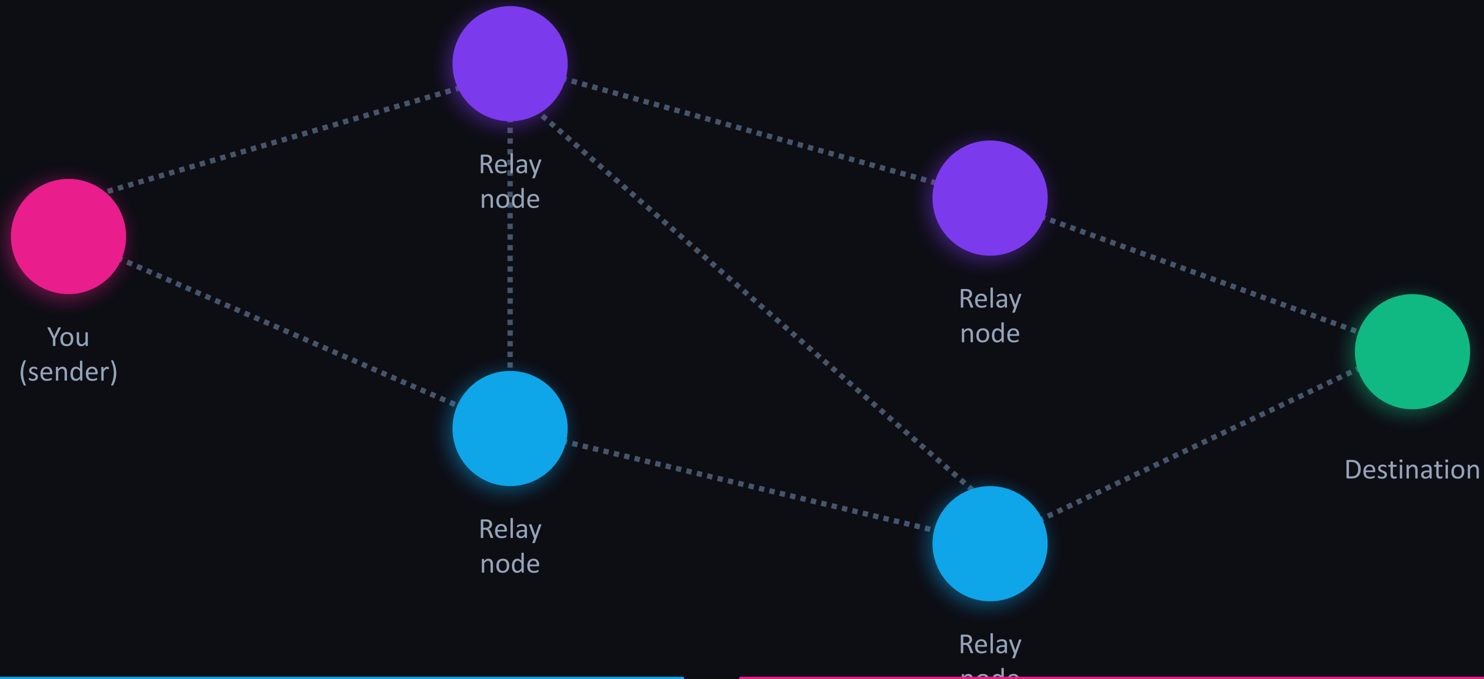
5 Configure your node info

Set a short name (4 chars) + long name. This is your identity on the mesh.

6 Send a test message

Type in the app → your node broadcasts → neighbors relay it across the mesh

How a Mesh Network Thinks



Flood routing: every node rebroadcasts messages it hasn't seen before. Hops limited by TTL (default: 3).

No message ever travels one path — the mesh finds all possible routes simultaneously.

Node Roles & Channels

NODE ROLES

CLIENT
Default. Sends messages, routes for neighbors, low power mode OK.

CLIENT_MUTE
Sends & receives but does NOT relay others' messages.

ROUTER
Optimized for high-traffic relay. Always awake, high power.

REPEATER
Only forwards, never generates messages. For fixed infrastructure.

CHANNELS

Primary channel
Where all devices must agree (name + PSK). LongFast is the default.

Secondary channels
Up to 7 additional channels per node. Each has its own PSK.

Admin channel
Remote management. Used to configure nodes without physical access.

Session 2

MQTT, Bridging & Designing for Resilience

→ MQTT protocol

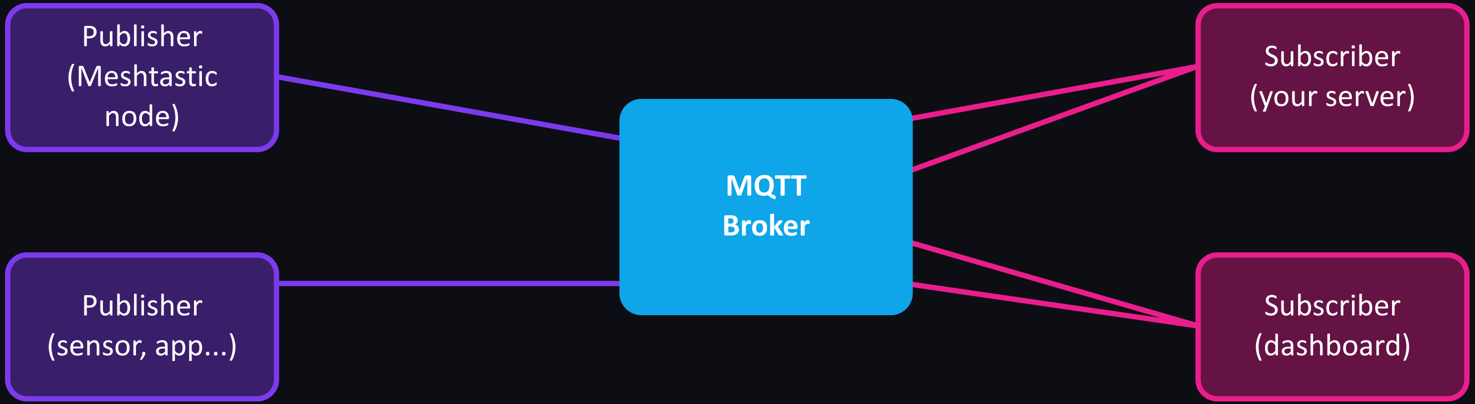
→ Bridging mesh to your web server

→ Designing resilient systems

What is MQTT?

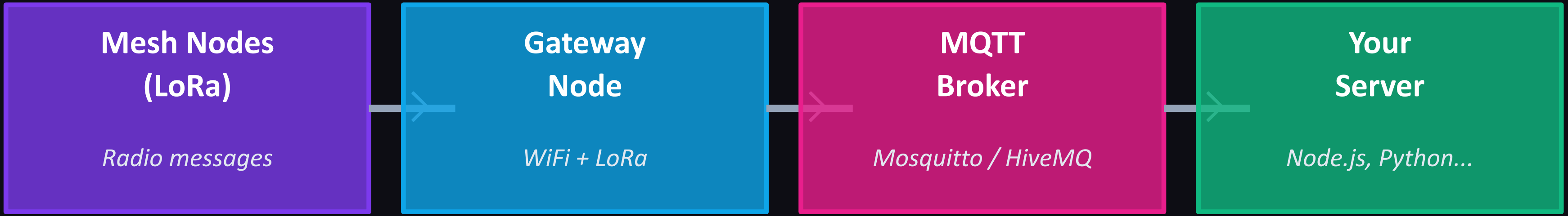
Message Queuing Telemetry Transport

A lightweight publish/subscribe messaging protocol. Think of it as a post office for machine messages — designed for low-bandwidth, unreliable networks.



`Topic example: msh/EU_868/2/json/+  — nodes publish to topics; subscribers filter by topic pattern`

Bridging Mesh → MQTT → Web



CONFIGURATION STEPS

1. On your Meshtastic node: Settings → MQTT → enable, set broker IP, port 1883, set topic prefix
2. Install Mosquitto (or use a hosted broker). Test with: mosquitto_sub -h localhost -t 'msh/#' -v
3. In your server (Python/Node): subscribe to the MQTT topic, parse the protobuf JSON payload
4. A gateway node needs WiFi access — it bridges the LoRa mesh to your IP network

What is Uplink / Downlink?

How your mesh node talks to the internet — and back.

Meshtastic device — **LoRa mesh** — **WiFi gateway node** — **MQTT broker** — **web app / server**

UPLINK — mesh → MQTT

The device publishes every message it hears on the LoRa mesh up to the MQTT broker. Any WiFi-connected node can do this — even if it didn't originate the message. Enable Uplink on a channel to activate it.

DOWNLINK — MQTT → mesh

The device subscribes to MQTT and broadcasts incoming messages onto the LoRa mesh. This is how you inject a message from a web server or script into a radio mesh. Enable Downlink on a channel to activate it.

Both can be enabled simultaneously — but if two devices both have uplink on, the broker gets duplicate messages. Deduplicate by message ID in your code.

Prerequisites

What you need before configuring MQTT on your node.

01
A Meshtastic device

Heltec WiFi LoRa 32, T-Beam, RAK WisBlock, or any ESP32-based LoRa board with WiFi support.

02
Meshtastic mobile app

Android or iOS. Used for all configuration via Bluetooth. Get it from the app store.

03
A shared WiFi network

The device and your laptop must be on the same network. The device uses WiFi to reach the MQTT broker.

04
An MQTT broker

We're using Mosquitto running on a Raspberry Pi at dweb2025.nohost.me, port 1883. No username or password required.

Step 1 — Connect to WiFi

Your node needs WiFi to reach the MQTT broker.

1	Connect to your device via Bluetooth in the Meshtastic app
2	Go to Radio Configuration → Network
3	Enable WiFi
4	Enter your WiFi SSID and password
5	Tap Save — the device will reboot and connect

After reboot, the device LED behavior may change. You can verify WiFi is connected via Radio Configuration → Network — it should show an IP address.

Step 2 — Create Channels

You need two channels: one for uplink, one for downlink.

Channel 0 is always primary — you can only have one. Channels 1–7 are secondary. Think walkie-talkie: Channel 0 is always on, the others are extras you monitor simultaneously.

Channel 0 (primary)

`broadcasts`

Uplink	ON
Downlink	off

Publishes messages from the mesh to MQTT. Both devices should have this on.

Channel 1 (secondary)

`mqtt`

Uplink	off
Downlink	ON

Subscribes to MQTT and broadcasts incoming messages to the mesh.

Messages published via MQTT to the mqtt channel topic will be received by devices with downlink enabled and rebroadcast to the mesh on the broadcasts channel.

Step 3 — MQTT Module Config

Module Configuration → MQTT in the Meshtastic app.

Enabled	**on**	
Server Address	**dweb2025.nohost.me**	
Username / Password	**(empty)**	*Our broker has no auth*
Encryption Enabled	**off**	*Must be off for JSON mode*
JSON Enabled	**on**	*Required for readable messages*
TLS Enabled	**off**	*Not needed on local network*
Root Topic	**msh/afterhours**	*Custom topic keeps traffic separate*
Proxy to Client Enabled	**off**	*Only needed if no device WiFi*

Step 4 — Configure LoRa Settings

Radio Configuration → LoRa in the Meshtastic app.

`Ignore MQTT`

`false`

If true, the device ignores any messages that originated from MQTT. Set to false so your mesh relays MQTT-injected messages normally.

`OK to MQTT`

`true`

Marks the device's messages as safe to forward to MQTT. If false, other nodes won't uplink this device's messages to the broker.

Both settings should be set on both devices. Without them, messages may be silently dropped or not relayed across the mesh.

EU compliance note

With a 6 dBi antenna in the EU, set TX Power to 8 dBm to stay within the legal EIRP limit of 14 dBm. Radio Configuration → LoRa → TX Power.

Step 5 — Verify with mosquitto

Install the mosquitto client tools and subscribe to your broker.

Install mosquitto client tools on your laptop:

Mac	`brew install mosquitto`
Windows (Chocolatey)	`choco install mosquitto`
Windows (installer)	`mosquitto.org/download`

Subscribe to all messages on your broker:

```
mosquitto_sub -h dweb2025.nohost.me -p 1883 -t 'msh/afterhours/#' -v
```

Send a message from the Meshtastic app on the broadcasts channel — you should see it appear in your terminal as JSON.

Test downlink (MQTT → mesh):

```
mosquitto_pub -h dweb2025.nohost.me -p 1883 -t 'msh/afterhours/2/json/mqtt/!NODE_ID' -m
'{"from":NODE_DECIMAL_ID,"type":"sendtext","payload":"hello"}'
```

MQTT Topic Structure

How Meshtastic organises messages in MQTT.

Every message is published to a topic with this structure:

```
{root_topic} / 2 / {format} / {channel} / {node_id}
```

`{root_topic}`	**msh/afterhours**	*Set in MQTT config*
`{format}`	**json or e**	*json = readable, e = protobuf binary*
`{channel}`	**broadcasts**	*Channel name from your config*
`{node_id}`	**!a6965508**	*The gateway node's hex ID*

Example (full uplink topic):

```
msh/afterhours/2/json/broadcasts/!a6965508
```

Who Uses This & Why It Matters

Disaster & Emergency

- → Puerto Rico after Hurricane Maria
- → Turkey/Syria earthquake response 2023
- → Wildfire evacuation coordination

Activism & Protest

- → Hong Kong protests (Bridgefy)
- → Surveillance-free coordination
- → Off-grid communication in repressive contexts

Remote & Rural

- → Hiking trail waypoint sharing
- → Farm sensor networks
- → Mountain rescue coordination

Art & Research

- → Distributed media installations
- → Urban sensing & data art
- → Community infrastructure projects

Key Takeaways

- LoRa = long range, low power, low bandwidth radio — perfect for off-grid text & sensor data.
- Meshtastic turns cheap hardware into a self-healing, encrypted mesh network with no infrastructure needed.
- Flood routing means there's no master, no central point of failure — the network thinks collectively.
- MQTT bridges your mesh to the internet. One WiFi-connected node is all you need as a gateway.
- Resilient design means planning for failure: redundancy, solar power, height, and degraded-mode testing.
- This is artistic territory. Invisible infrastructure made visible. Networks as a medium.